

"Predicting Rainfall Using Machine Learning Classification Algorithms"

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Agenda

Problem

Literature Review

Contribution

Significance

Methodology

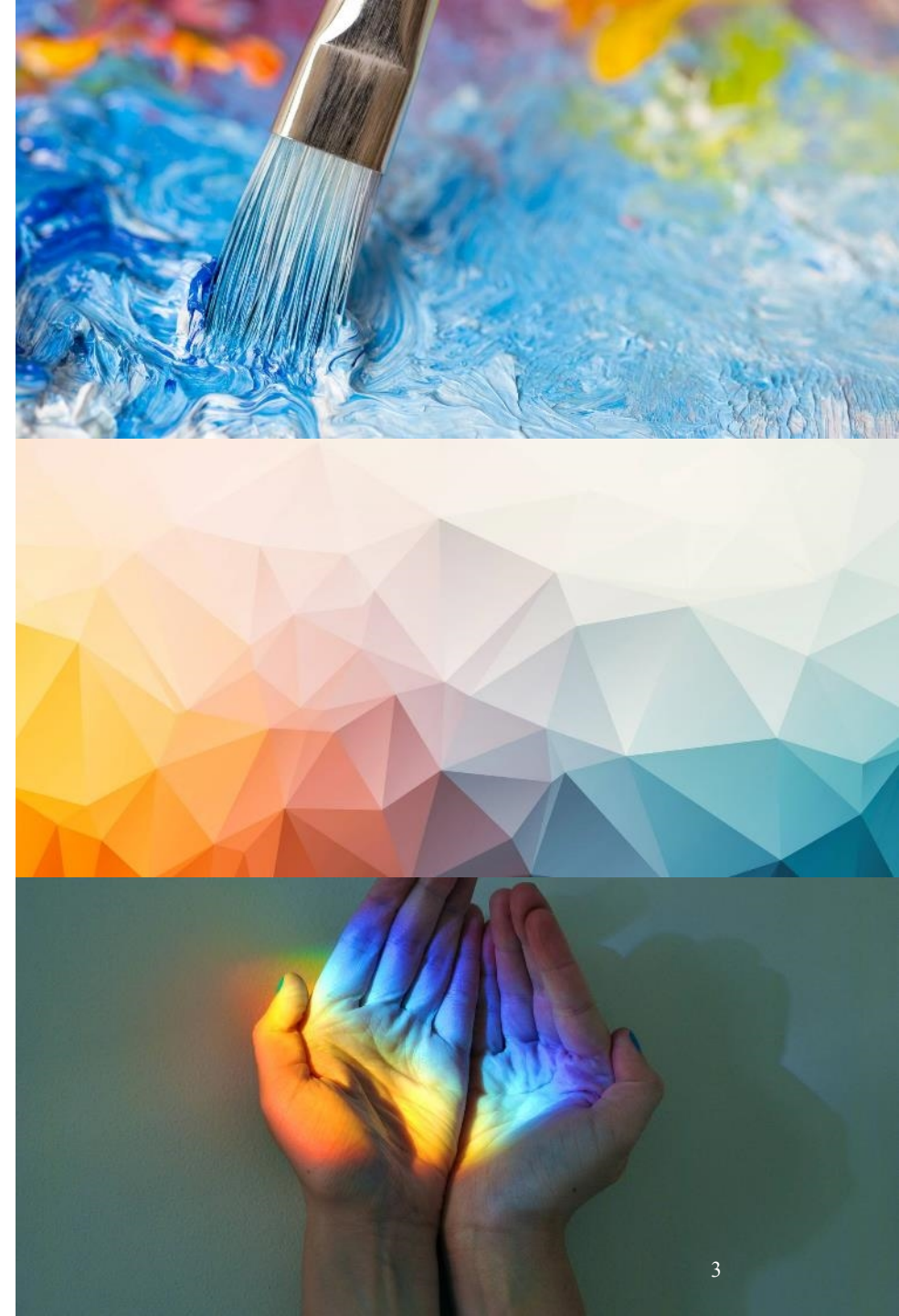
Algorithm



Problem Statement

Project Overview: Predicting whether it will rain tomorrow using historical weather data from Australia.

Importance: Accurate weather predictions help with global weather forecasting and disaster preparedness.



Literature Review:

Importance of Rainfall Prediction: Critical for regions like Australia, where diverse climatic conditions and water resource management are crucial.

Limitations of Traditional Models: Previous research highlights that statistical models fail to capture nonlinear relationships in meteorological data.

Machine Learning Advantage: Algorithms like Random Forest, Gradient Boosting, and Neural Networks outperform traditional models by handling complex datasets and variable interactions.

Efficiency of Hybrid Models: Research incorporating Neural Networks, Random Forest, and K-Nearest Neighbors shows improved prediction accuracy with large-scale data.

Our Approach: Expanding on prior work by including Logistic Regression alongside machine learning algorithms for enhanced comparative analysis and predictive performance.

Contributions:

Application of Machine Learning: Demonstrates the practical use of machine learning concepts in real-world rainfall prediction.

Algorithm Optimization: Focuses on data preprocessing, feature selection, and comparing model accuracy to optimize predictions.

Effective Model Implementation: Highlights the effectiveness of Decision Tree, KNN, Logistic Regression, and other models in handling diverse meteorological variables.

Practical Framework: Contributes a scalable, data-driven framework for rainfall prediction applicable to various regions.

Reinforced Knowledge: Reinforces understanding of machine learning concepts by applying them to a real-world challenge.

Significance

Crucial for Key Sectors: Supports accurate decision-making in agriculture, disaster management, and water resource planning.

Enhanced Resource Allocation: Improves resource allocation by predicting rainfall across diverse climatic zones.

Urban Pollution Mitigation: Helps in predicting rainfall's natural cleansing effect on air quality, aiding pollution management.

Timely Action for Public Health: Enables timely measures to reduce pollution in smog-prone areas, improving public health.

Global Relevance: Predictive models can be adapted to other regions, enhancing global environmental management strategies.

EDA

Importing libraries and loading dataset :

NumPy, pandas, matplotlib, seaborn

Dataset: Kaggle

Data Exploration: Shape & Head, Dataset Info, Summary statistics

Data Preprocessing: Categorical Encoding- The script converts the Rain Today and Rain Tomorrow categorical columns into binary values.

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 142193 entries, 0 to 142192
Data columns (total 24 columns):
 #   Column              Non-Null Count  Dtype  
---  -
 0   Date                142193 non-null object  
 1   Location            142193 non-null object  
 2   MinTemp             141556 non-null float64 
 3   MaxTemp             141871 non-null float64 
 4   Rainfall            140787 non-null float64 
 5   Evaporation         81350 non-null  float64 
 6   Sunshine            74377 non-null  float64 
 7   WindGustDir         132863 non-null object  
 8   WindGustSpeed       132923 non-null float64 
 9   WindDir9am          132180 non-null object  
10   WindDir3pm          138415 non-null object  
11   WindSpeed9am        140845 non-null float64 
12   WindSpeed3pm        139563 non-null float64 
13   Humidity9am         140419 non-null float64 
14   Humidity3pm         138583 non-null float64 
15   Pressure9am         128179 non-null float64 
16   Pressure3pm         128212 non-null float64 
17   Cloud9am            88536 non-null  float64 
18   Cloud3pm            85099 non-null  float64 
19   Temp9am             141289 non-null float64 
20   Temp3pm             139467 non-null float64 
21   RainToday           140787 non-null object  
22   RISK_MM             142193 non-null float64 
23   RainTomorrow        142193 non-null object  
dtypes: float64(17), object(7)
memory usage: 26.0+ MB
```

Methodology

Objective: Briefly explain that you are solving a classification problem to predict whether it will rain tomorrow based on weather data.

Handling Imbalanced Datasets: Techniques to balance the data, so the model doesn't favor the majority class over the minority class.

Converting Categorical Data to Numbers: Turning non-numeric data (like categories) into numbers so machine learning models can understand them.

Advanced Methods for Missing Data: Using methods like Multiple Imputation by Chained Equations (MICE) to fill in missing data more accurately.

Identifying and Removing Outliers: Finding and removing extreme values that can distort the model's predictions.

Choosing the Best Features: Using methods to pick the most important features (variables) to improve model performance.

Comparing Model Performance: We tested 5 different algorithms on both unbalanced and balanced datasets, and then compared their performance.

Logistics Regression

What it does:

Logistic Regression predicts the likelihood of an event happening, like whether it will rain tomorrow (Yes/No). Think of it as a yes-or-no decision-maker.

How it works:

- It starts by combining weather features (e.g., temperature, humidity) into a single equation.
- This value is passed through a special function called the sigmoid function, which squishes the output into a probability between 0 and 1.
- If the probability is greater than 0.5, the model predicts "Yes" (Rain), otherwise "No" (No Rain).

Why use it here?

Logistic regression is simple, interpretable, and efficient for binary classification problems like this one.

Decision Tree

What it does:

A Decision Tree is like a flowchart where each step asks a yes-or-no question about the weather to narrow down whether it will rain.

How it works:

- Each "branch" of the tree splits the data based on a weather feature (e.g., "Is Humidity > 70%?").
- The process continues until it reaches a final decision (Rain or No Rain).

Why use it here?

Decision Trees are easy to visualize and understand, making them great for explaining predictions.

Random Forest

What it does:

Random Forest is like a team of decision-makers (trees) that vote on whether it will rain tomorrow. Each tree gives its prediction, and the majority wins.

How it works:

- It builds many decision trees using different parts of the data.
- Each tree predicts Rain/No Rain based on specific weather features.
- All trees vote, and their combined decision is the final prediction.

Why use it here?

Random Forest is powerful because it reduces errors caused by individual trees and works well with complex relationships in data.

KNN

What it does:

KNN predicts based on the similarity of today's weather to past days. It's like asking, "When we had similar weather before, did it rain the next day?"

How it works:

- The algorithm finds the "k" most similar weather days in the dataset.
- It checks how many of those days had Rain or No Rain.
- The prediction is based on the majority outcome.

Why use it here?

KNN is simple yet effective when patterns in the data are clear.

Neural Network

What it does:

Neural Networks mimic how our brains work by learning patterns in data. It's like having a weather expert analyze multiple complex factors to decide if it will rain tomorrow.

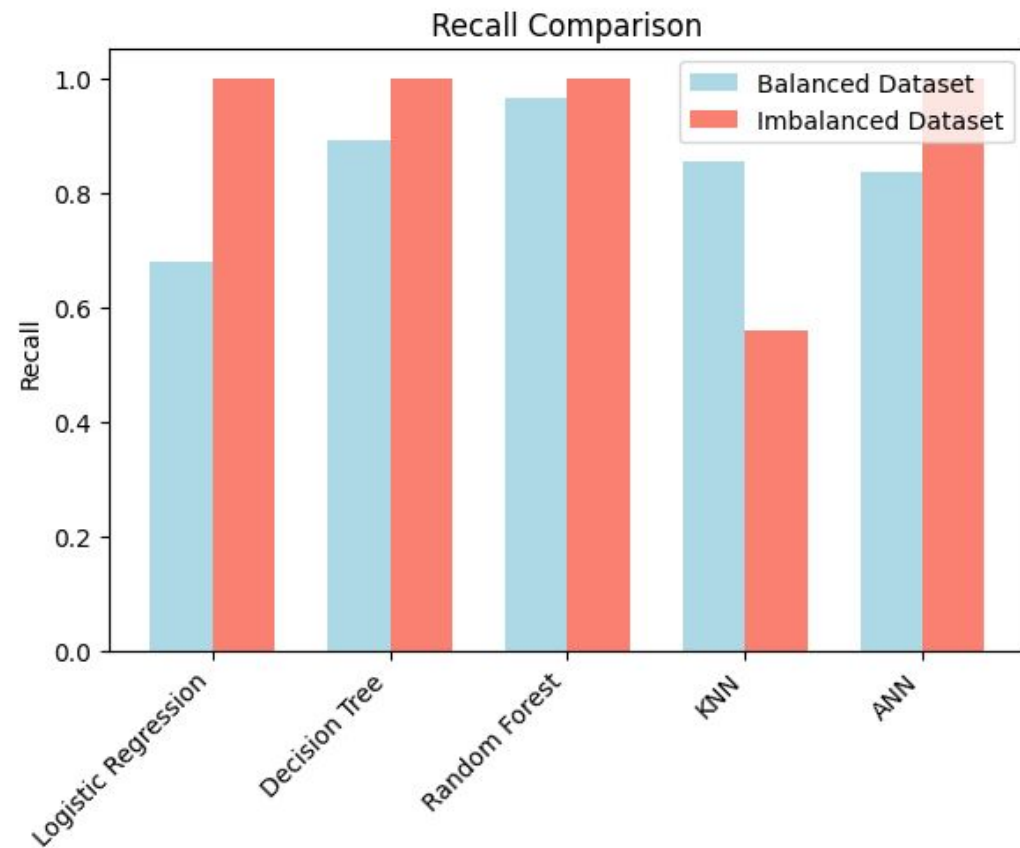
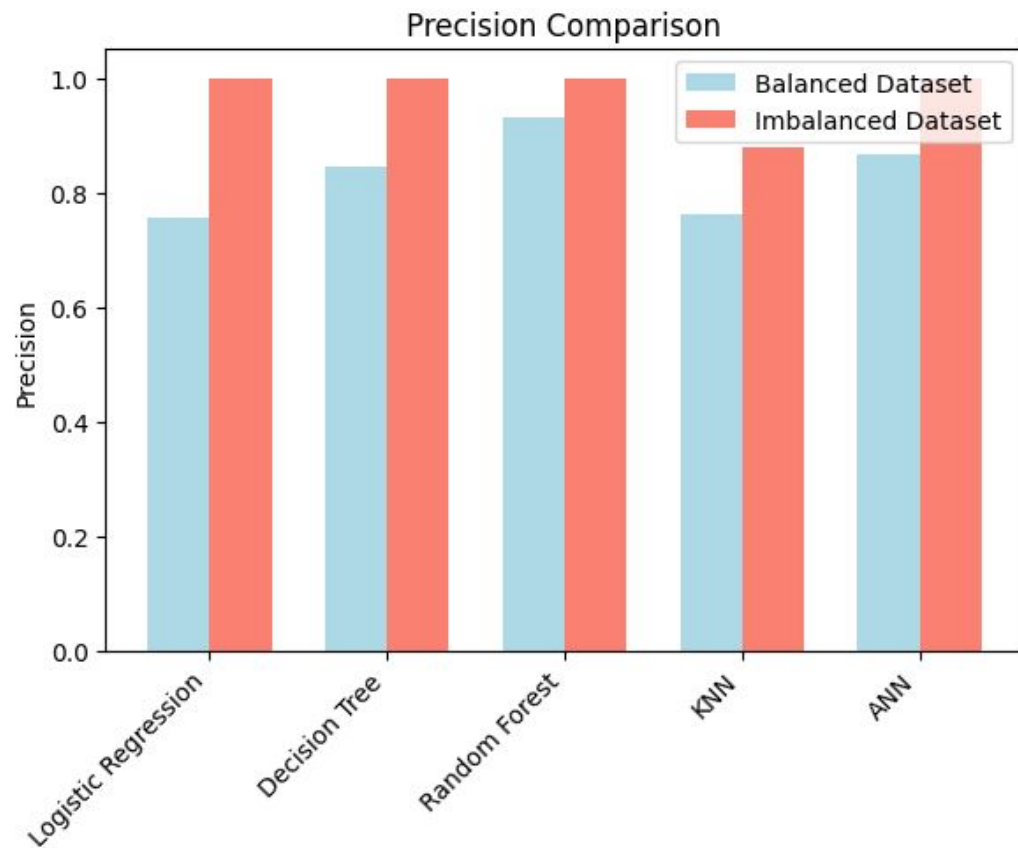
How it works:

- The data flows through layers of interconnected nodes (neurons).
- Each layer learns to identify patterns, like "High humidity + Low temperature = Rain."
- These layers combine to make a final prediction.

Why use it here?

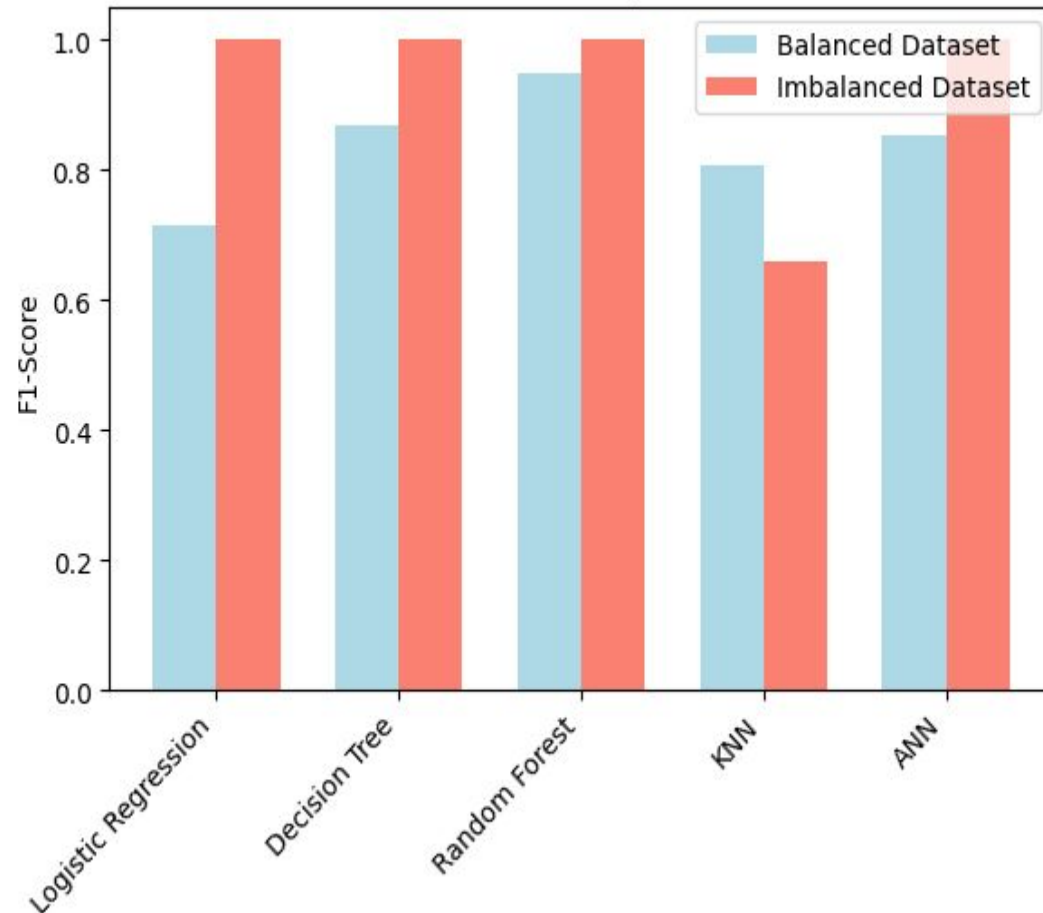
Neural Networks are excellent for capturing subtle relationships in large, complex datasets.

Evaluation metrics

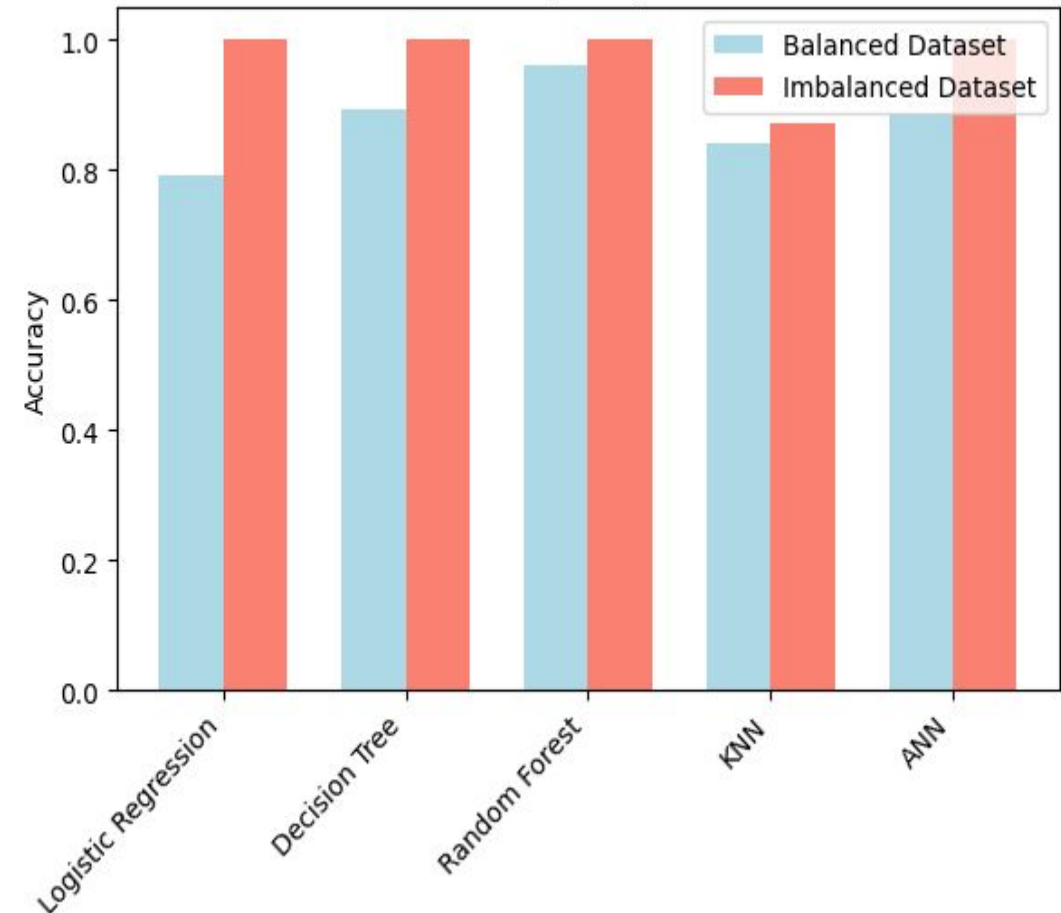


Evaluation metrics

F1-Score Comparison



Accuracy Comparison



Model Performance: Balanced vs Imbalanced Datasets

1. Balanced Dataset:

- Random Forest performs the best with high accuracy, precision, recall, and F1-score.
- ANN and Logistic Regression are slightly lower than Random Forest.
- KNN and Decision Tree show decent performance, with KNN having lower recall.

2. Imbalanced Dataset:

- Random Forest and ANN handle imbalance well with high precision, recall, and F1-score.
- KNN struggles with a significant drop in recall for the minority class.

Why Models Differ:

- Random Forest and ANN handle complex patterns and imbalances effectively.
- Logistic Regression and Decision Trees are more biased toward the majority class.
- KNN is prone to majority class bias due to its reliance on proximity.



Any Questions ?

Thank you